

Himani Mistry

Netid: ls3844

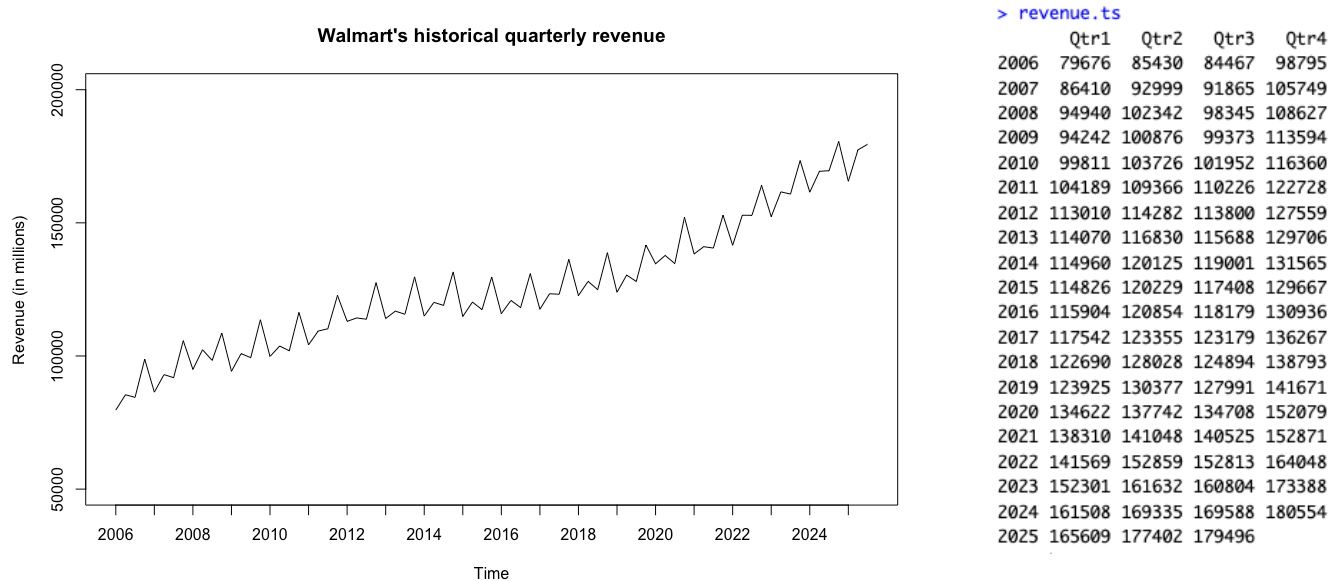
BAN673

February 22, 2026

Supporting documents: *Himanicase2.R file*

Case Study 2: Walmart Revenue Forecasting

Question #1



A time series data set was created using the provided quarterly revenue data from Walmart - named *revenue.ts*. This data was plotted to visualize the historical trends. The time series shows a strong upward trend in Walmart's quarterly revenue over time. There is clear seasonal behavior, with revenues consistently peaking in Q4 each year, likely due to holiday sales. The increasing slope in recent years suggests possible nonlinear growth.

Question #2

The data set was then partitioned to create a validation partition (19 periods or quarters in this case) and a training partition (the rest of the historical data).

```
nValid <- 19
nTrain <- length(revenue.ts) - nValid

train.ts <- window(revenue.ts, end = time(revenue.ts)[nTrain])
valid.ts <- window(revenue.ts, start = time(revenue.ts)[nTrain + 1])
```

Three regression models were applied to the training partition of this data set. Here are the following summaries:

Regression model with seasonality

```
tslm(formula = train.ts ~ season)
```

Residuals:

Min	1Q	Median	3Q	Max
-29045	-11575	4080	9012	26473

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	108721	3768	28.854	< 2e-16 ***
season2	5050	5329	0.948	0.34740
season3	3351	5329	0.629	0.53205
season4	16885	5329	3.169	0.00248 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 14590 on 56 degrees of freedom

Multiple R-squared: 0.1693, Adjusted R-squared: 0.1248

F-statistic: 3.803 on 3 and 56 DF, p-value: 0.01493

Model equation: Revenue = 108721 + 5050Dq2 + 3351 Dq3 + 16885 Dq4

The seasonal regression model estimates quarterly dummy variables using Q1 as the reference category. The coefficient for Q4 is statistically significant, indicating that revenue in Q4 is significantly higher than in Q1. However, the coefficients for Q2 and Q3 are not statistically significant. The model explains approximately 16.9% of the variation in revenue (based on R^2), indicating a relatively weak fit. Although the overall F-test is statistically significant ($p = 0.01493$), suggesting that at least one seasonal effect is different from zero, the model does not include a time trend. As a result, it fails to capture the upward growth in revenue over time, and the forecasts repeat the same seasonal pattern each year without growth.

Validation period forecast:

```
> model1_pred <- forecast(model_season, h = nValid, level = 0)
```

```
> model1_pred
```

	Point Forecast	Lo 0	Hi 0
2021 Q1	108721.1	108721.1	108721.1
2021 Q2	113770.7	113770.7	113770.7
2021 Q3	112071.7	112071.7	112071.7
2021 Q4	125606.4	125606.4	125606.4
2022 Q1	108721.1	108721.1	108721.1
2022 Q2	113770.7	113770.7	113770.7
2022 Q3	112071.7	112071.7	112071.7
2022 Q4	125606.4	125606.4	125606.4
2023 Q1	108721.1	108721.1	108721.1
2023 Q2	113770.7	113770.7	113770.7
2023 Q3	112071.7	112071.7	112071.7
2023 Q4	125606.4	125606.4	125606.4
2024 Q1	108721.1	108721.1	108721.1
2024 Q2	113770.7	113770.7	113770.7
2024 Q3	112071.7	112071.7	112071.7
2024 Q4	125606.4	125606.4	125606.4
2025 Q1	108721.1	108721.1	108721.1
2025 Q2	113770.7	113770.7	113770.7
2025 Q3	112071.7	112071.7	112071.7

Regression model with linear trend and seasonality

```
tslm(formula = train.ts ~ trend + season)
```

Residuals:

Min	1Q	Median	3Q	Max
-6812.5	-2446.3	-381.9	3060.1	7465.0

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	85694.43	1138.94	75.241	< 2e-16 ***
trend	794.02	25.24	31.455	< 2e-16 ***
season2	4255.58	1234.17	3.448	0.00109 **
season3	1762.55	1234.95	1.427	0.15917
season4	14503.19	1236.23	11.732	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3379 on 55 degrees of freedom

Multiple R-squared: 0.9563, Adjusted R-squared: 0.9531

F-statistic: 300.6 on 4 and 55 DF, p-value: < 2.2e-16

Model equation: Revenue = 85694.43 + 794.02T + 4255.58Dq2 + 1762.55Dq3 + 14503.19 Dq4

The linear trend with seasonality model includes both a time trend and quarterly dummy variables. The trend coefficient (794.02) is highly statistically significant, indicating that Walmart's revenue increases by approximately \$794 million per quarter. Q2 and Q4 are statistically significant seasonal effects, while Q3 is not. The model explains 95.63% of the variation in revenue ($R^2 = 0.9563$), indicating a very strong fit. The overall model is highly statistically significant ($p < 2.2e-16$). By incorporating both growth and seasonality, this model produces increasing forecasts over time and better reflects the underlying revenue pattern.

Validation period forecast:

```
> model2_pred <- forecast(model_linear, h = nValid, level = 0)
> model2_pred
```

	Point	Forecast	Lo 0	Hi 0
2021	Q1	134129.9	134129.9	134129.9
2021	Q2	139179.5	139179.5	139179.5
2021	Q3	137480.5	137480.5	137480.5
2021	Q4	151015.2	151015.2	151015.2
2022	Q1	137306.0	137306.0	137306.0
2022	Q2	142355.6	142355.6	142355.6
2022	Q3	140656.6	140656.6	140656.6
2022	Q4	154191.3	154191.3	154191.3
2023	Q1	140482.1	140482.1	140482.1
2023	Q2	145531.7	145531.7	145531.7
2023	Q3	143832.7	143832.7	143832.7
2023	Q4	157367.4	157367.4	157367.4
2024	Q1	143658.2	143658.2	143658.2
2024	Q2	148707.8	148707.8	148707.8
2024	Q3	147008.8	147008.8	147008.8
2024	Q4	160543.5	160543.5	160543.5
2025	Q1	146834.3	146834.3	146834.3
2025	Q2	151883.9	151883.9	151883.9
2025	Q3	150184.9	150184.9	150184.9

Regression model with quadratic trend and seasonality

```
tslm(formula = train.ts ~ trend + I(trend^2) + season)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-4914.3	-1858.3	-600.4	1937.2	7673.1

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	81899.443	1338.764	61.175	< 2e-16 ***
trend	1161.865	89.516	12.979	< 2e-16 ***
I(trend^2)	-6.030	1.422	-4.240	8.8e-05 ***
season2	4243.516	1078.843	3.933	0.000241 ***
season3	1750.491	1079.520	1.622	0.110726
season4	14503.194	1080.643	13.421	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2954 on 54 degrees of freedom

Multiple R-squared: 0.9672, Adjusted R-squared: 0.9641

F-statistic: 318.3 on 5 and 54 DF, p-value: < 2.2e-16

Model equation: Revenue = 81899.443 + 1161.865T - 6.030T² + 4243.516D_{Q2} + 1750.491D_{Q3} + 14503.194D_{Q4}

The quadratic trend with seasonality model includes both a linear and quadratic time trend, along with quarterly dummy variables. The linear trend term is positive and statistically significant, while the quadratic term is negative and significant, indicating that revenue increases over time but at a gradually decreasing rate. Q2 and Q4 seasonal effects are statistically significant, while Q3 is not. The model explains 96.72% of the variation in revenue ($R^2 = 0.9672$), which is slightly higher than the linear trend model. The overall model is highly statistically significant ($p < 2.2e-16$). By incorporating nonlinear growth and seasonal effects, this model provides the strongest statistical fit among the regression models considered.

Validation period forecast:

```
> model3_pred <- forecast(model_quad, h = nValid, level = 0)
```

```
> model3_pred
```

	Point Forecast	Lo 0	Hi 0
2021 Q1	130334.9	130334.9	130334.9
2021 Q2	134998.6	134998.6	134998.6
2021 Q3	132913.7	132913.7	132913.7
2021 Q4	146062.4	146062.4	146062.4
2022 Q1	131943.2	131943.2	131943.2
2022 Q2	136558.6	136558.6	136558.6
2022 Q3	134425.4	134425.4	134425.4
2022 Q4	147525.9	147525.9	147525.9
2023 Q1	133358.4	133358.4	133358.4
2023 Q2	137925.6	137925.6	137925.6
2023 Q3	135744.2	135744.2	135744.2
2023 Q4	148796.5	148796.5	148796.5
2024 Q1	134580.8	134580.8	134580.8
2024 Q2	139099.7	139099.7	139099.7
2024 Q3	136870.1	136870.1	136870.1
2024 Q4	149874.1	149874.1	149874.1
2025 Q1	135610.1	135610.1	135610.1
2025 Q2	140080.8	140080.8	140080.8
2025 Q3	137802.9	137802.9	137802.9

Comparing accuracy measures:

```
> round(accuracy(model1_pred$mean, valid.ts), 3)
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set 45285.07 46902.59 45285.07 27.968 27.968 0.828    4.926
> round(accuracy(model2_pred$mean, valid.ts), 3)
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set 13858.43 16006.32 13858.43  8.356  8.356 0.795    1.66
> round(accuracy(model3_pred$mean, valid.ts), 3)
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set 21639.69 24071.78 21639.69 13.135 13.135 0.814    2.501
```

Which are the two most accurate regression models for forecasting?

Based on the validation accuracy measures, the linear trend with seasonality model has the lowest RMSE (16006.32) and MAPE (8.356), indicating the best forecasting performance. The quadratic trend model is the second most accurate, while the seasonal-only model performs significantly worse. Therefore, the linear and quadratic regression models are selected as the two most accurate models for forecasting.

Question #3

The linear trend with seasonality model, and the quadratic trend model were then applied to the whole data set to make a forecast into the future. Here are the models developed for the full set and their summaries:

Regression model with linear trend and seasonality (full set)

```
tslm(formula = revenue.ts ~ trend + season)
```

Residuals:

Min	1Q	Median	3Q	Max
-10299	-5671	1115	3363	17077

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	80296.70	1798.54	44.645	< 2e-16 ***
trend	1005.36	30.09	33.413	< 2e-16 ***
season2	4930.79	1927.79	2.558	0.0126 *
season3	2698.68	1928.50	1.399	0.1659
season4	13960.36	1952.99	7.148	5.2e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6095 on 74 degrees of freedom

Multiple R-squared: 0.9409, Adjusted R-squared: 0.9377

F-statistic: 294.4 on 4 and 74 DF, p-value: < 2.2e-16

Model equation: $80296.70 + 1005.36T + 4930.79D_2 + 2698.68D_3 + 13960.36D_4$

The trend coefficient (1005.36) is positive and highly statistically significant ($p < 2e-16$), indicating that Walmart's revenue increases by approximately \$1005 million per quarter. Q2 and Q4 seasonal effects are statistically significant, while Q3 is not. The model explains 94.09% of the variation in revenue ($R^2 = 0.9409$), indicating a strong overall fit. The F-test confirms that the model is statistically significant overall ($p < 2.2e-16$). Therefore, this model is appropriate for forecasting.

Regression model with quadratic trend and seasonality (full set)

```
tslm(formula = revenue.ts ~ trend + I(trend^2) + season)
```

Residuals:

Min	1Q	Median	3Q	Max
-9115.8	-4109.7	299.7	3893.7	9914.2

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	88385.286	1931.622	45.757	< 2e-16	***
trend	398.904	98.478	4.051	0.000126	***
I(trend^2)	7.581	1.193	6.355	1.60e-08	***
season2	4938.372	1557.392	3.171	0.002222	**
season3	2698.682	1557.961	1.732	0.087463	.
season4	14362.138	1579.012	9.096	1.23e-13	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4924 on 73 degrees of freedom

Multiple R-squared: 0.9619, Adjusted R-squared: 0.9593

F-statistic: 369 on 5 and 73 DF, p-value: < 2.2e-16

Model equation: $88385.286 + 398.904T + 7.581T^2 + 4938.372D_{Q2} + 2698.682D_{Q3} + 14362.138D_{Q4}$

The quadratic trend and seasonal model includes both linear and quadratic time components, allowing revenue growth to accelerate over time. Both the trend and quadratic terms are statistically significant, indicating increasing growth over time. Q2 and Q4 seasonal effects are statistically significant, while Q3 is not. The model explains 96.19% of the variation in revenue ($R^2 = 0.9619$), indicating an excellent fit. The overall F-test confirms the model is statistically significant ($p < 2.2e-16$). Therefore, this model is suitable for forecasting.

Future forecast comparison: Quarter 4 2025, 4 Quarters of 2026, 4 Quarters of 2027

```
> future9.df
```

	Linear_Forecast	Quadratic_Forecast
1	174685.8	183176.1
2	161730.8	170433.4
3	167666.9	177006.3
4	166440.2	176416.4
5	178707.2	189744.7
6	165752.2	177062.6
7	171688.4	183696.2
8	170461.6	183166.8
9	182728.6	196555.8

Because the quadratic T is positive in the full-dataset model, the quadratic forecasts grow at an increasing rate and are consistently higher than the linear forecasts in later quarters.

Comparing accuracy measures: (include naive and seasonal naive)

```
> round(accuracy(naive_full$fitted, revenue.ts), 3)
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set 1279.744 9796.452 8260.795 0.721 6.677 -0.718      1
> round(accuracy(snaive_full$fitted, revenue.ts), 3)
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set 4729.24 5882.997 4883.613 3.652 3.789 0.735      0.555
> round(accuracy(model_linear_full$fitted, revenue.ts), 3)
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set 0 5899.425 4850.481 -0.115 3.699 0.887      0.541
> round(accuracy(model_quad_full$fitted, revenue.ts), 3)
      ME      RMSE      MAE      MPE      MAPE      ACF1 Theil's U
Test set 0 4733.615 4033.899 -0.15 3.328 0.863      0.484
```

Which forecast is more accurate?

Based on RMSE and MAPE, the quadratic trend with seasonality model provides the most accurate forecasts. It has the lowest RMSE (4033.899) and lowest MAPE (3.328%), outperforming the linear trend model and both naive benchmarks. Therefore, the quadratic model is the most appropriate model to forecast Walmart's quarterly revenue for Q4 2025 and Q1–Q4 of 2026–2027.